management and school administration, to library management. Mostly, the systems listed are quite popular, with a good development and user community, and the software are quite stable and feature rich. Systems like Moodle have a large national and international user base. Many of these are also available in multiple Indian and international languages.

Future outlook

We can see that there is a strong synergy between the open movements and academic education. There is a lot that open movements are bringing and *can* further bring to academic programmes. We need to encourage our academia to benefit from this and also contribute back to help the growth of the movements, in return. Our own projects, students' projects, as well as PhD/MS projects can benefit tremendously from the existing resources, and can be used to drive new developments and significant enhancements. This needs to happen on a larger scale.

At the same time, there are new challenges coming up on the education side. Content sharing across institutions has happened relatively less often, so far. But with the growth of e-content and the increasing presence of institutions on the Web, this is bound to occur more frequently. Inter-institutional collaboration in sharing not only content, but full courses, subjects and faculty is certainly possible soon. This will necessitate a lot of changes in the software requirements, and offers good opportunities for us to contribute and also adapt existing software to meet these new requirements. New demands will also be made on interoperability as records and resources move across institutional boundaries. Initiatives such as OKI are a step in that direction. Work on distributed learning management systems also looks at similar concerns.

The sharing, in turn, also brings into focus the growing concern of plagiarism. With a vast collection of resources freely available, the chances of plagiarism and the difficulty in detecting it, is increasing. Scigen is a relevant case study, which produces 'scientific' research papers using some natural language processing techniques. Since the language appears of good quality, rich with a high degree of relevant jargon and following the style and conventions of a research paper, it appears genuine, and outputs from this program have been accepted in some international conferences. Tracking copied (with and without distortion) submissions for assignments to research papers is a major challenge in the academic environment.

There are also changes at the libraries. Good quality open source solutions are available to handle the functionalities of today's libraries. Even digital libraries are well supported by FOSS solutions such as Dspace. However, with the growth of e-content rich with simulations, and e-learning growing in popularity, the nature of resources that the library needs to deal with is changing. Dealing with IPSEs offers different challenges compared to conventional or even digital books.

Various FOSS applications by category

Application category	FOSS applications available
Web browser	Firefox, Iceweasel, Konqueror, Epiphany
Document creation	OpenOffice.org, KOffice, Latex
Audio record/edit	Audacity, Ardour
Web page creation	Nvu, Bluefish, Quanta Plus
Content management	Drupal, Jhoomla, Plone/Zope
Learning management	Moodle, Sakai, Atutor
Question banking, testing	exe2learn, Moodle
School administration	schooltool
Visual programming	scratch
Diagram editing	Dia
Scanner	Xsane
3D animation	Blender
Image editing	The GIMP, Krita
Page layout program	Scribus
Plotting	Kmplot
Creating and running tests	Keduca
Video conferencing	Dimdim, Vmukti, Ekiga, open-meetings
Library management	Koha, Dspace

Table 2

Already, issues of subscription management and sharing of e-resources is a major concern.

We also need to look at removing the linguistic and physical barriers preventing people from using technology. Software localisation and accessibility are two fields related to these two aspects, which need to see a lot more activity, for countries like India.

In summary, FOSS has enriched the education field in many ways. But the world is moving fast in the education sector as well as other sectors, and new demands and opportunities are constantly emerging on the horizon. FOSS needs to be sustained and nurtured through a sustained cycle of human resources and efforts, to help it continue what it has been able to do so far. END

References

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- UNESCO portal on FOSS offering a list of open source resources as content and general software of use in education: *www.unesco-ci.org/cgi-bin/portals/foss/page.cgi?d=1&g=10*
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